B.TECH DATA SCIENCE

Module 1: Introduction to Data Science

B.Tech Data Science — Expanded Study Notes and Exam Preparation

Contents
Four chapters of expanded concepts, definitions, examples, applications, methods and revision questions.
Includes side headings on every study page and important questions at the end for exam preparation.

Prepared for learning and exam revision. Follow your university syllabus and faculty guidance where they differ.

1.1 Foundations of Data Science

Study focus: Introduction to Data Science. Read the side headings, understand the example, and revise the key ideas.

Definition

Data science is an interdisciplinary field that combines statistics, programming, analytical reasoning and subject knowledge to extract useful information from data. It helps people describe events, understand causes, predict outcomes and make evidence-informed decisions.

Why it matters

Organisations collect data from transactions, websites, sensors, surveys, machines and public records. Data science converts these records into understandable evidence. For example, a college may analyse attendance and assessment records to identify students who need academic support.

Core ingredients

The main ingredients are data, statistical reasoning, computing, domain knowledge and communication. Data without context can be misleading; a model without evaluation can be unreliable; and a correct result that is not communicated clearly may not be useful.

Data-to-decision path

A typical path begins with a question, gathers relevant data, checks quality, explores patterns, builds an analysis or model, evaluates the result and communicates a recommendation. Monitoring is needed when the result is used repeatedly.

Example

A retail store asks why sales fell. Analysts compare sales by date, product, location, price and stock availability. They may discover that a popular product was unavailable on weekends. The insight supports an operational change rather than a guess.

1.2 Goals and Types of Analysis

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Descriptive analysis

Descriptive analysis answers “What happened?” It summarises past observations using counts, totals, averages, percentages, tables and charts. A monthly attendance report is a descriptive analysis.

Diagnostic analysis

Diagnostic analysis asks “Why might it have happened?” It compares groups, examines relationships and investigates possible explanations. A drop in attendance may be associated with timetable changes, transport problems or other factors; association alone does not prove a cause.

Predictive analysis

Predictive analysis estimates what may happen using patterns in historical data. Examples include predicting demand, forecasting electricity use and estimating the risk of late delivery. Predictions are uncertain and should be evaluated on data not used to fit the model.

Prescriptive analysis

Prescriptive analysis considers actions that may improve an outcome, often using constraints, optimisation or simulation. A delivery service may compare routes under limits on distance, time and vehicle capacity.

Choosing the goal

The analytical goal determines the data needed, the method, the success measure and the final communication. A clearly written question prevents a project from becoming an unfocused search for interesting patterns.

1.3 Related Fields

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Data analytics

Data analytics often focuses on examining data to answer practical questions, create reports and explain performance. Its methods may include SQL queries, summary statistics, dashboards and experiments.

Data engineering

Data engineering designs and maintains the systems that collect, store, transform and deliver data reliably. Tasks include building pipelines, managing databases, checking data quality and controlling access.

Artificial intelligence

Artificial intelligence (AI) is a broad area concerned with systems that perform tasks associated with intelligent behaviour. Machine learning is one approach within AI that learns patterns from examples.

Machine learning

Machine learning develops algorithms that learn patterns from data for tasks such as classification, regression, clustering and recommendation. It is useful when a rule-based solution is difficult to write manually.

Business intelligence

Business intelligence uses data, reporting and dashboards to monitor organisational performance. The fields overlap: a data scientist may use an engineered data pipeline, statistical analysis and a dashboard to deliver a result.

1.4 Data Science Workflow

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Step 1 — Problem definition

Identify the stakeholder, the decision to support, the target outcome, constraints and a measurable definition of success. “Improve student outcomes” is broad; “identify students who may need support before the next assessment” is more actionable.

Step 2 — Data collection

Choose data sources that are relevant, permitted and sufficiently representative. Record the population, sampling method, time period, units, definitions and any consent or access requirements.

Step 3 — Cleaning and exploration

Check missing values, duplicates, inconsistent units, invalid records and unusual values. Use summary statistics and charts to understand distributions, trends and possible data problems.

Step 4 — Feature preparation and modelling

Convert raw records into useful variables, choose a suitable method and establish a simple baseline. Split data appropriately before fitting transformations or models to avoid information leakage.

Step 5 — Evaluation and communication

Evaluate using measures that reflect the task, explain uncertainty and limitations, and present the result in language appropriate for the audience. If deployed, monitor performance and data changes.

1.5 Iteration, Stakeholders and Success Measures

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Iterative process

Data science is rarely a perfectly straight line. Evaluation may reveal that labels are unreliable, the question is poorly defined or a model is not useful. The team may return to earlier steps and revise the data or approach.

Stakeholders

Stakeholders include the people who request the analysis, the people represented by the data, technical teams, decision-makers and those affected by the resulting action. Their needs and risks may differ.

Success measures

Success measures should be defined before model selection. Examples include prediction error, time saved, cost reduction, service availability or the proportion of cases correctly identified. A technically accurate model may still fail to help the real decision.

Constraints

Common constraints include limited time, budget, computing capacity, privacy requirements, data access, fairness concerns and the need to explain decisions.

Documentation

Record the question, data sources, cleaning decisions, assumptions, model version, evaluation results and known limitations. Documentation makes work easier to reproduce, review and maintain.

2.1 Types of Data by Structure

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Structured data

Structured data follows a defined schema, commonly rows and columns. Examples include student tables, sales records and inventory databases. Its regular structure makes filtering, joining and aggregation straightforward.

Semi-structured data

Semi-structured data has organisational markers but may not fit a fixed table. JSON objects and XML documents use keys or tags to describe fields. Different records can contain different attributes.

Unstructured data

Unstructured data does not naturally follow a simple row-and-column schema. Examples include free text, photographs, audio, video and scanned documents. Processing may require language, image or signal techniques.

Choosing storage

The data structure affects storage, validation and analysis. A relational table is suitable for consistent transactions, while documents or object storage may be more convenient for variable records and media.

Practical caution

A file extension does not guarantee quality. A CSV can contain inconsistent dates, missing values or ambiguous column names. Every source still needs inspection and documentation.

2.2 Qualitative and Quantitative Data

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Qualitative data

Qualitative data describes categories or qualities. Examples include department, product type, colour and feedback category. Nominal categories have no natural order; ordinal categories have an order, such as low, medium and high.

Quantitative data

Quantitative data represents numerical quantities. Discrete data is countable, such as number of calls. Continuous data can take any value within a range, such as height, time or temperature.

Measurement scales

Nominal data identifies categories; ordinal data adds rank; interval data has meaningful differences but no true zero; ratio data has meaningful differences and a true zero. The scale affects which calculations are sensible.

Units and definitions

Numbers without units can be misleading. A value of 12 could mean seconds, kilograms or items. Data dictionaries should document units, permitted values and definitions.

Example

Student branch is nominal, satisfaction rating may be ordinal, number of absences is discrete, and time spent on a task is continuous. Treating categories as ordinary measurements can produce meaningless averages.

2.3 Primary and Secondary Data Sources

Study focus: Introduction to Data Science. Read the side headings, understand the example, and revise the key ideas.

Primary data

Primary data is collected directly for the current purpose through surveys, interviews, experiments, observations or sensors. The collector can design questions and measurements around the research question.

Secondary data

Secondary data already exists, such as public datasets, research reports, administrative records or archived transaction data. It may save time but may have been collected for a different purpose.

Internal and external sources

Internal sources include company databases, attendance records and operational logs. External sources include public statistics, weather records and published research. Combining sources requires compatible definitions and permissions.

Sampling and bias

A sample should represent the target population. Convenience sampling may over-represent easy-to-reach participants. Non-response and poorly worded questions can also distort results.

Data provenance

Provenance records where data came from, when it was collected, who transformed it and how it was used. It supports trust, reproducibility and appropriate attribution.

2.4 Data Tables and Data Dictionaries

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Rows and columns

A row usually represents an observation or entity, while a column represents a variable. A well-designed table states what one row means and avoids mixing multiple kinds of observations without a clear key.

Identifiers

A unique identifier distinguishes records. Names are often poor identifiers because they may be repeated or changed. When combining tables, keys must be checked for uniqueness and consistent meaning.

Data dictionary

A data dictionary documents field name, description, data type, units, allowed values, missing-value conventions and examples. It helps different team members interpret data consistently.

Schema validation

Schema checks detect unexpected columns, incorrect types, invalid categories and missing required fields. Validation should happen near ingestion so errors do not silently spread.

Example

For a student table, `student_id` may be a string identifier, `attendance_pct` a percentage from 0 to 100, and `semester` a defined category. These rules make quality checks possible.

3.1 Roles in a Data Science Team

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Data analyst

A data analyst typically answers questions with queries, summary statistics, reports and visualisations. The emphasis is often on explaining what the data shows and helping stakeholders make decisions.

Data scientist

A data scientist may design experiments, perform statistical analysis, build predictive models and evaluate whether patterns generalise beyond the observed data.

Data engineer

A data engineer builds and maintains data pipelines, storage, processing systems and quality controls so analysis can use reliable data.

Domain expert and product owner

Domain experts explain the meaning of variables and operational realities. Product owners or decision-makers clarify priorities and decide how findings should be used.

Collaboration

Responsibilities overlap across organisations. Effective work requires shared definitions, clear handoffs, version control, peer review and communication about uncertainty.

3.2 Common Tools and Technologies

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Spreadsheets

Spreadsheets are useful for small datasets, quick inspection, formulas, charts and communicating results. They become difficult to govern when many people edit copies or when data is very large.

SQL and databases

SQL is used to select, filter, join, group and aggregate relational data. Databases provide structured storage and controlled access to records.

Python and R

Python offers libraries for data manipulation, visualisation and machine learning. R is widely used for statistical analysis and graphics. Tool selection depends on team skills, project needs and the environment.

Notebooks and version control

Notebooks combine code, output and explanation, which is useful for exploration. Git records changes to code and documents, allowing teams to review and reproduce work.

Visualisation and platforms

Dashboard tools present metrics interactively. Cloud and distributed platforms can support larger workloads, but introduce costs, security responsibilities and operational complexity.

3.3 Ethics, Privacy and Security

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Privacy and consent

Collect only data needed for a legitimate purpose. Inform people appropriately, follow relevant rules and limit access. Removing names alone may not fully anonymise a dataset if other fields can identify a person.

Security

Protect data with access controls, secure storage, encryption where appropriate, backups and careful sharing. Sensitive data should not be placed in public repositories or unsecured files.

Bias and representation

Historical data can reflect unequal access, measurement errors or past discrimination. A model trained on unrepresentative records may perform poorly for groups missing from the data.

Transparency

Explain the purpose, inputs, limitations and consequences of an analysis. High-impact decisions may need stronger review, human oversight and clear ways to challenge errors.

Responsible reporting

Do not exaggerate findings or hide inconvenient results. State assumptions, uncertainty and limitations, and avoid implying that correlation proves causation.

4.1 Communicating Findings

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Audience first

A technical audience may need model details and validation results, while a decision-maker may need the recommendation, expected impact, cost and risks. Adapt the presentation without changing the evidence.

Story structure

A clear report introduces the question, describes the data and method, presents key findings, explains uncertainty and limitations, and finishes with practical next steps.

Evidence and uncertainty

Report sample size, time period and relevant uncertainty. Avoid presenting estimates as exact facts. Distinguish observed results from interpretations and recommendations.

Visual clarity

Use descriptive titles, readable labels, consistent units and appropriate scales. Remove decoration that distracts from the message, and ensure the chart can be understood without guessing.

Reproducibility

Include enough information about sources, transformations and methods for a reviewer to understand how the conclusion was reached.

4.2 Example Case Study: Student Support

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Problem

A university wants to identify students who may benefit from earlier academic support. The aim is to offer assistance, not to label students permanently or automatically punish them.

Data and preparation

Potential fields may include attendance, assessment completion and prior course results, subject to appropriate permissions. Check missingness, inconsistent records and whether every field is available at the intended prediction time.

Analysis

Begin with descriptive summaries and discuss possible explanations with academic staff. If a prediction model is used, compare it with a simple baseline and evaluate on separate data.

Ethical safeguards

Restrict access, explain how the output will be used, examine performance across relevant groups and provide human review. Avoid unnecessary sensitive variables and monitor unintended consequences.

Evaluation

Assess whether the approach helps students receive timely support, not only whether a metric improves. Collect feedback, document limitations and revise the process when evidence changes.

4.3 Reproducible and Maintainable Work

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Organised project

Separate raw data, processed data, code, reports and outputs. Keep original data unchanged when possible, and make transformations explicit.

Version control

Use version control to track changes to scripts and documentation. Meaningful commit messages and code reviews help identify when and why a result changed.

Environment and dependencies

Record the software versions and libraries required to run an analysis. Different versions can change behaviour or numerical results.

Testing

Check functions on typical cases, boundary cases and invalid inputs. Validate row counts, ranges and expected totals after important transformations.

Audit trail

Record the dataset version, code version, assumptions, evaluation results and date of analysis. This supports reproducibility and responsible investigation.

5. Important Exam Questions and Revision

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Long-answer questions

1. Define data science and explain its scope. 2. Explain the data science life cycle with a diagram. 3. Differentiate structured, semi-structured and unstructured data with examples. 4. Discuss roles of data analyst, data scientist and data engineer. 5. Explain ethical issues in data science.

Short-answer questions

1. What is descriptive analysis? 2. Define data provenance. 3. What is a data dictionary? 4. Distinguish discrete and continuous data. 5. List four data science tools. 6. What is data leakage? 7. Define bias in data collection.

Case-based questions

1. Design a workflow for analysing student attendance. 2. Explain how you would check a public dataset before using it. 3. Describe privacy safeguards for a healthcare dataset. 4. Explain how to communicate uncertain results to a non-technical manager.

Revision checklist

Be able to define data science, describe the workflow, classify data types, compare related roles, identify quality and privacy risks, and explain how results are communicated and monitored.

Exam tip

For a 5- or 10-mark answer, start with a definition, use side headings, explain each point, add a simple example or diagram, and finish with a short conclusion.